**Subject:** Mathematics | **WEEKLY CHAPTER TEST – 01** | **Time:** 45 Minutes
**Class:** Class X (CBSE) | **Chapter 6: Triangles** | **MM: 20**

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory. Marks are indicated against each question.
2. **Section A** consists of 5 questions of 1 mark each (4 MCQs and 1 Assertion-Reason).
3. **Section B** consists of 4 questions of 11 marks total (1 VSAQ of 2 marks and 3 SA of 3 marks each).
4. **Section C** consists of 1 Case-Based integrated unit of assessment of 4 marks with sub-parts.
5. Use of calculators is strictly prohibited. Neat geometric sketches must accompany geometric proofs.

## SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each)

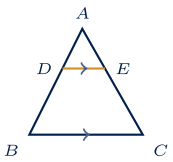
**1.** Two polygons of the same number of sides are similar if:

(a) Their corresponding angles are equal and corresponding sides are equal
(b) Their corresponding angles are proportional and sides are equal
(c) Their corresponding angles are equal and corresponding sides are proportional
(d) Their perimeters and areas are equal

**2.** Which of the following geometric figures are **always** similar to each other?

(a) Two rectangles (b) Two rhombuses (c) Two equilateral triangles (d) Two isosceles triangles

**3.** In $\Delta ABC$, $D$ and $E$ are points on sides $AB$ and $AC$ respectively such that $DE \parallel BC$. If $AD = 3.6$ cm, $AB = 9.6$ cm, and $AE = 2.4$ cm, then the length of $AC$ is:




(a) 6.4 cm (b) 5.4 cm (c) 4.0 cm (d) 7.2 cm

**4.** In $\Delta ABC$, $DE \parallel BC$ such that $AD = x$, $DB = x - 2$, $AE = x + 2$, and $EC = x - 1$. The value of $x$ is:

(a) 3 (b) 4 (c) 5 (d) 6

**5.** **Assertion (A):** Two congruent figures are always similar.
**Reason (R):** Similar figures always have the same size and the same shape.

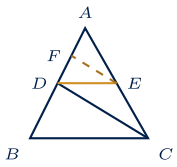
(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

## SECTION B: DESCRIPTIVE QUESTIONS (PART 1)

**6.** **[2 Marks]** In $\Delta ABC$, $D$ and $E$ are points on sides $AB$ and $AC$ respectively. If $AD = 5.7$ cm, $BD = 9.5$ cm, $AE = 3.3$ cm, and $AC = 8.8$ cm, determine whether $DE \parallel BC$, giving mathematical justification.

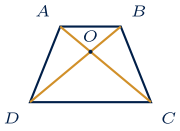
**7.** **[3 Marks]** State Basic Proportionality Theorem (Thales' Theorem). Using this theorem, prove that a line drawn through the midpoint of one side of a triangle parallel to another side bisects the third side.

## SECTION B: DESCRIPTIVE QUESTIONS (PART 2 Continued)




**8.** **[3 Marks]** In the given figure, $DE \parallel BC$ and $CD \parallel EF$. Prove mathematically that:

$$AD^2 = AB \times AF$$

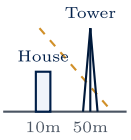



**9.** **[3 Marks]** In a trapezium $ABCD$ with $AB \parallel DC$, diagonals $AC$ and $BD$ intersect each other at the point $O$. Using the concept of similarity of triangles or Basic Proportionality Theorem, show that:

$$\frac{AO}{OC} = \frac{BO}{OD}$$

## SECTION C: CASE STUDY BASED QUESTION (4 Marks)

**10.** **Case Study 01: Shadow Triangulation of Vijay's House and City Tower**




Vijay is trying to find the height of a transmission tower near his house using the properties of similar triangles. The height of Vijay's house is 20 m. At a specific time of day, Vijay's house casts a shadow 10 m long on the ground, while at the same time the transmission tower casts a shadow 50 m long, and his neighbour Ajay's house casts a shadow 20 m long.

(i) **[1 Mark]** What is the height of the transmission tower?

(ii) **[1 Mark]** What is the height of Ajay's house?

(iii) **[2 Marks]** When the transmission tower casts a shadow of 40 m, what will be the length of the shadow cast by Vijay's house at that instant?

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